The Role of Diet in Mother-Infant Reciprocity in the Spiny Mouse

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One-day-old spiny mouse pups responded preferentially to bedding soiled by lactating conspecifics fed the same diet as their mothers. Following this test, pups were fostered onto different-diet females. When retested at 84-96 hr of age, no preferences were shown for bedding soiled by a female fed the biological mother's diet vs bedding of a female fed the different diet. When tested again at 120-132 hr of age, however, the pups preferred the bedding associated with their foster mothers' diet. In a 2nd experiment, recently parturient females retrieved 1-day-old pups born of same-diet females faster than pups born to novel-diet females. These results indicate that pup preferences for chemical cues produced by lactating conspecifics can be altered by sufficient exposure to a 2nd female maintained on a different diet and that neonatal chemical cues, like maternal chemical stimuli, are diet-dependent.

Leon (1975) demonstrated that laboratory rat pups are attracted to the maternal chemical cues emitted by lactating rats on the same diet as the pups' own mothers. The pups exhibited no preferences for different-diet maternal cues. Furthermore, related research with highly precocial spiny mouse (*Acomys cahirinus*) infants has revealed that whereas maternal chemical cues produced by this species appear to possess species-specific characteristics, these may be less salient than dietary factors in affecting responsiveness of the neonates (Porter, Deni, & Doane, in press; Porter & Doane, in press). Specifically, the Porter and Doane (in press) study employed 2 species of rodents, *A. cahirinus* and *Mus musculus*. Regardless of species, *A. cahirinus* pups were attracted to the chemical cues emitted by lactating females maintained on the same diet as the pups' biological mother.

Given the dietary-dependent nature of maternal chemical cues, and the likelihood of rapid changes in diet in the wild, one might expect that *A. cahirinus* neonates would show altered responsiveness to differing chemical complexes as a function of changes in exposure to those stimuli. However, a recent study (Porter & Etscorn, 1975) using relatively brief exposure to arbitrarily selected olfactory stimuli (spices) found that 2-day-old *A. cahirinus* pups respond preferentially to the chemical stimulus to which they had been exposed for 1 hr on the day of birth (Day 1) as compared to

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a 2nd stimulus which they had experienced for a comparable length of time on Day 2. When the length of exposure to the Day 2 odor was 2 or 4 times as long as the length of exposure to the Day 1 odor, no preferences were subsequently found for either stimulus. Therefore, a "primacy" effect appears to occur following equal periods of exposure during the 1st 2 days of life, but this effect is not immutable for it can be precluded (but not reversed) by increasing the relative length of exposure to the 2nd (recent) stimulus. This somewhat surprising failure to find a complete reversal of the primacy effect may be to some extent an artifact of the arbitrary chemical stimuli as well as the length and context of the exposure to those stimuli. Accordingly, the present study was designed to test further the malleability of responsiveness of *A. cahirinus* pups to chemical stimuli by using prolonged exposure to cues emanating from lactating females maintained on 2 distinct diets.

Experiment I

Subjects

All subject animals were the offspring of spiny mice (*Acomys cahirinus*) maintained in breeding pairs. Six weeks prior to the beginning of data collection, the colony was divided and housed in 2 different rooms. Both rooms were kept on a fixed lighting schedule (on at 0600 hours and off at 1900 hours) and at the same temperature (23°C ± 2°) and relative humidity (50%). The parent colony remained on Purina Mouse Chow whereas the secondary colony was fed a test diet (previously used by Leon [1975] for related research with rats) supplied by Teklad Test Diets under the code number TD 75342. (The primary ingredients are potato flour, casein, cellulose, and sucrose.) The diet of both groups was supplemented with a small amount of carrot 4 days a week and a tablespoon of oats twice weekly. A total of 47 pups, 22 (8 litters) of which were the offspring of Purina-fed females and 25 (10 litters) from Teklad-fed females, comprised the subject pool.

Procedure

Newly born pups were allowed to remain with their natural mother 24-30 hr. During this period, lactating (2-11 days postpartum) stimulus females from each diet colony were individually isolated in clean cages supplied with fresh sugar cane waste as bedding material. Each sequestered stimulus female remained in her respective colony room for 22-26 hr and was provided only water ad lib. In no case was the stimulus female either the biological or foster mother of any pup tested.

For testing purposes, a shallow layer of visibly soiled bedding material from each stimulus cage was placed on opposite sides of a cage measuring 28 x 18 x 13 cm. A clean center strip, 4 cm wide, separated the 2 bedding layers. Testing was performed in a room where no animals were housed or food stored. The choice test consisted, in all cases, of bedding soiled by two 2-11 day postpartum females, one fed Purina chow vs one fed the Teklad diet.

Each pup was tested individually with a maximum of 4 pups from 1 litter serving as subjects. As a pup was removed from its home cage, it was placed into a holding cage whose floor was covered with a clean paper towel. The pup was then positioned in the center of the median strip always facing the back wall at a 90° angle to the bedding. Each pup was given 7 trials of this 2-choice test. As soon as the pup was placed within the testing chamber, a stopwatch was activated. It was stopped when either a choice had been made (i.e., all 4 paws placed on the bedding) or a 2-min interval had elapsed. At the end of each trial, the pup was removed and placed in the holding cage while the test chamber was rotated 180°. No more than 1 litter was tested in the same cage.

Each pup was tested on 3 separate days, with the 1st test occurring when the pups were 24-36 hr old. Immediately after all members of a litter were tested for the 1st time, they were fostered onto different-diet females, e.g., pups whose biological mother was maintained on the Teklad diet were fostered onto a Purina-fed female. All foster mothers had given birth within 5 days of the parturition date of the biological mother. The pups were tested again using the aforementioned paradigm at 72-84 hr of age (4th day of life) and at 120-132 hr old (6th day of life). After each of these testing sessions, the pups were returned to their foster mothers.

In order to rule out any possible effect due to fostering, we tested a within-diet fostering control group. In this condition, a total of 20 pups from each diet colony were used as subjects. The testing environment and paradigm were exactly the same as previously described. However, these pups were fostered onto same-diet females (e.g., Teklad pups fostered onto Teklad females) at the appropriate time.

Results

A single preference score was obtained for each pup by subtracting the number of times it chose bedding soiled by a female on the novel diet from choices of bedding soiled by a female on the diet of its biological mother. Accordingly, a positive score indicates that the subject preferred the bedding associated with a female on the biological mother's diet whereas a negative score indicates a preference for the novel-diet bedding. The mean preference scores within each diet were computed separately. Because the direction and the magnitude of these scores were essentially the same, data from both the Purina and Teklad diet were combined for each of the 3 testing ages in both the cross-diet and within-diet groups. (The number of subjects in the experimental condition was reduced for each successive test. This was due to an unusually high rate of cannibalism which occurred after fostering the pups onto different-diet females. That is, of the 46 experimental pups fostered into different diet females, 37 survived until the 3rd test session. Within the control group, however, all 20 pups fostered onto the same-diet females were alive for the final test.)

The control subjects across the testing sessions preferred the chemical cues associated with their biological mother's diet. (See Table 1.) Experimental subjects initially preferred the chemical cues associated with the diet of the biological mother, whereas on the 2nd test day, they exhibited no preference for either stimulus. However by the 3rd testing session, they exhibited a significant preference for the chemical cues associated with the diet of their foster mothers. The mean response latencies in seconds for the experimental pups were 12.77 (1st test), 18.74 (2nd test), and 9.67 (3rd test). An analysis of variance revealed that these means were

TABLE 1. Summary of Results of Choice Tests Between Bedding Soiled by Lactating Females Maintained on the Maternal vs Novel Diet.

	Experimental Group (Fostered onto Novel Diet Female Following Test 1)				Control Group (Fostered onto Maternal Diet Female Following Test 1)			
	Mean Preference Scores[a]				Mean Preferences Scores			
Test	Infants Born to Purina Females	Infants Born to Teklad Females	n	Total	Infants Born to Purina Females	Infants Born to Teklad Females	n	Total
1 (24-36 hr)	+ 2.7	+ 3.4	46	+ 3.2[d]	+ 3.7	+ 2.3	20	+ 3.00[c]
2 (72-84 hr)	+ .7	− .1	46	+ .4	+ 2.9	+ 1.8	19	+ 2.35[c]
3 (120-132 hr)	− 2.1	− 3.1	35	− 2.6[d]	+ 3.3	+ 2.9	20	+ 3.1[d]

[a]+ Score = Preference for the bedding soiled by a lactating female maintained on the biological mother's diet. − Score = Preference for the bedding soiled by a lactating female maintained on the novel diet.

[b]Because the Wilcoxon matched-pairs signed-ranks test deletes tied scores from the analyses, the overall n may be greater than that presented in the table.

[c]$p < .01$.

[d]$p < .001$.

significantly different ($F = 9.28$, $df = 2/30$, $p < .001$). A post-hoc Neuman-Keuls analysis revealed that the means of the 1st and 3rd test latencies were not appreciably different from one another whereas the mean latencies of the 2nd session were significantly longer than either of the other sessions. The mean latencies of the control group in seconds were: 1st test, 12.24; 2nd, 13.59; and 3rd, 9.70. The analysis of variance here yielded no significant differences ($F = 2.65$, $df = 2/12$, $p > .05$).

Experiment II

As mentioned above, the cannibalism rate was considerably higher for the cross-diet (Experimental) as compared to the within-diet (Control) fostering condition. These data therefore suggest that chemical cues associated with neonates, like those emanating from lactating females, may be dietary-dependent, and that recently parturient females discriminate between (and respond differentially towards) pups born of females maintained on the same diet as themselves as opposed to different-diet females. The role of dietary factors in maternal responsiveness to strange offspring was investigated further in Experiment II.

Subjects

A total of 24 multiparious *A. cahirinus* females served as subjects for Experiment II. Thirteen of these females were from the Purina-diet colony (see details above) whereas the remaining 11 were from the Teklad-diet colony. As in Experiment I, 2 colony rooms were used to house separately the animals in the 2 dietary conditions.

Apparatus

The test case was a galvanized steel enclosure measuring 66 x 46 x 20 cm high divided into 2 sections (46 x 30 and 46 x 36 cm, respectively) by a metal partition. A 10 x 18-cm high opening centered at the bottom of the partition allowed for movement between the 2 compartments during testing. Prior to the test session, animals were confined in the smaller section of the enclosure (isolation chamber) by placing a barrier of bricks against the opening in the partition. The floor of the isolation chamber was lined with bedding material (sugar-cane waste).

Procedure

Within 60 hr postpartum, individual lactating females, along with their mates and offspring, were removed from their home cage and placed into the isolation chamber of the testing enclosure. A separate testing enclosure was used for females in each of the dietary conditions; testing was conducted in each subject's own colony room.

Approximately 24 hr after being confined in the isolation chamber, each female was tested for her responses to two 24-36 hr-old conspecific pups with one of these pups born to a female from the Teklad-diet colony and the 2nd born to a Purina-diet female. One hour prior to testing, all animals other than the female to be tested (i.e., adult male and offspring) were removed from the isolation chamber. At the time of testing, the 2 stimulus infants (1 from each of the dietary conditions) were placed onto the center of the bare floor in the 46 x 36-cm retrieval area adjacent to the isolation chamber containing the subject female. The barrier at the opening between the retrieval area and isolation chamber was then reduced to a single brick set on its broad surface. This 6-cm high barrier could be easily surmounted by the adult female, but posed a major obstacle to the stimulus pups. Each female was then observed for a maximum of 15 min or until both neonates were retrieved (i.e., picked up by the female, carried across the barrier, and deposited into the isolation chamber). The order and latencies of retrieval of the 2 pups were recorded.

Results

Six of the 24 lactating females (3 each from the Teklad and Purina-diet conditions) showed no retrieval during their 15-min test sessions. Therefore, the analysis of the results of Experiment II was based upon the 18 females who each retrieved at least 1 of their 2 stimulus pups.

Of the 10 Purina-diet females who displayed retrieval behavior, all 10 retrieved their respective Purina-diet pup whereas 8 also retrieved the Teklad-diet pup. Mean retrieval latencies for the 10 Purina females were 470.3 sec for the Purina pups and 614.9 for the Teklad pups. In the Teklad group, all 8 retrieving females retrieved their respective Teklad-diet pups, and 7 of the Purina pups were retrieved. Mean retrieval latencies (by the Teklad-diet females) of Teklad- vs Purina-diet pups were 377.3 and 439.6 sec, respectively. The statistical analysis (t-test for correlated samples) of females' own diet vs novel diet, for all 18 retrieving females, revealed that the retrieval latencies for own-diet pups ($\bar{X}$ = 428.9) was significantly less than that for the novel-diet pups ($\bar{X}$ = 537.0; t = 3.02, df = 17, $p < .01$; 2-tailed test).

Discussion

The data reported in Experiment I indicate that the attachment of *A. cahirinus* pups to specific dietary-dependent maternal chemical cues is malleable. That is, when tested on the 2nd day of life, the neonates significantly preferred bedding soiled by a female maintained on the same diet as their mother (with whom they had been housed prior to testing) over bedding soiled by a novel-diet lactating female. This preference for chemical cues associated with the maternal diet was not irreversible, however, in that cross-diet animals tested on the 6th day of life, following 4 days of rearing by a foster female maintained on a diet different from that of the pups' mother, preferred chemical cues emanating from lactating females maintained on the latter diet. The within-diet fostered animals, however, showed a consistent preference for bedding soiled by females fed the same diet as their biological and foster mothers over bedding soiled by lactating females maintained on a novel diet.

Such a reversal in preferences by the cross-diet condition for dietary-dependent maternal chemical cues appears to require a considerable period of exposure to the "novel" chemical stimuli in that the test following 2 days of rearing with a foster mother was followed by non-preferential responding to chemical cues produced by a different-diet vs maternal-diet lactating females. Thus, the failure of Porter and Etscorn (1975) to find a reversal of the primacy effect in their earlier study may have been a function of the relatively brief duration of their exposure sessions.

The response latencies exhibited by the cross-diet pups in the 2nd test session were significantly greater than for those of either the 1st or the final tests. This is in keeping with the lack of overall preference for either complex of chemical stimuli in this same test condition. During the 2nd test, the cross-diet pups behaved as if "indecisive" in the face of 2 equally attractive chemical cues. A maturation effect can be ruled out as a possible explanation for this latency discrepancy because the within-diet pups did not exhibit increased response latencies during the 2nd test relative to the other 2 test sessions.

Porter and Etscorn, in their studies (1974, 1975) of olfactory attachment formed by *A. cahirinus* pups to artificial odorants, have noted the possible similarity between these olfactory attachments and visual imprinting in birds. However, although the stimulus complex (i.e., both visual and auditory components) to which the precocial avian hatchlings become imprinted remain relatively constant over time, dietary-dependent pheromones might be expected to change (along with the available food items) within the nursing period of a given litter of *A. cahirinus*. This hypothesis is strengthened by the recent finding that precocial *A. cahirinus* pups show a prolonged period of suckling (extending through at least the 30th day of life) in a laboratory setting (R. H. Porter and H. M. Doane, unpublished observation). Thus, in order to maintain proximity with the mother, *Acomys* pups should be able to modify their olfactory attachments according to changing environmental factors.

The data presented by Porter and Doane (in press) combined with the present research clearly demonstrate that odors emitted by an *A. cahirinus* female during lactation are altered as a function of qualitative dietary change. More important, when this evidence is examined along with that of Leon (1975) for rats, it suggests that these odors are not species-specific and that diet may be implicated in group recognition.

Retrieval of neonates in *A. cahirinus* (R. H. Porter and H. M. Doane, unpublished observation) has been characterized as a prolonged (up to 30 days postpartum) and nonspecific with the same diet pups as long as the *A. cahirinus* females are lactating. The preferential retrieval by recently parturient females of own-diet vs novel-diet neonates (Experiment II) indicates that chemical cues associated with infant *A. cahirinus* may likewise vary as a function of maternal diet. These cues might be produced by the neonates themselves, be deposited on the surface of the neonates by their mother, or arise from ingestion of mother's milk. Some evidence exists to favor the latter possibility in rodents. Galef and Sherry (1973) and Bronstein, Levine, and Marcus (1975) have demonstrated that the mother's milk in rats contains chemical cues which later direct the type and quantity of food eaten by weanlings. This information is nongenetic as such in that the young are capable of acquiring these food habits from foster mothers maintained on radically dissimilar diets from the biological mother.

Although the females did respond preferentially to own-diet offspring, they still behaved maternally to the majority of the novel-diet pups. Therefore, despite the increased level of cannibalism of the between-diet vs within-diet pups in Experiment I following fostering and the ability of recently parturient *A. cahirinus* females to discriminate between pups as a function of dietary novelty (Experiment II), lactating females appear to be accepting of neonates born of females maintained on a strange diet.

Notes

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